

Overview of Machine translation

Subject: Machine translation

1.

1960–1975 Researchers continued to join the field as the Association for Machine Translation and Computational Linguistics was formed in the U.S. (1962) and the National Academy of Sciences formed the Automatic Language Processing Advisory Committee (ALPAC) to study MT (1964). Real progress was much slower, however, and after the ALPAC report (1966), which found that the ten-year-long research had failed to fulfill expectations, funding was greatly reduced. According to a 1972 report by the Director of Defense Research and Engineering (DDR&E), the feasibility of large-scale MT was reestablished by the success of the Logos MT system in translating military manuals into Vietnamese during that conflict. The French Textile Institute also used MT to translate abstracts from and into French, English, German and Spanish (1970); Brigham Young University started a project to translate Mormon texts by automated translation (1971).

2.

A deep learning-based approach to MT, neural machine translation has made rapid progress in recent years. However, the current consensus is that the so-called human parity achieved is not real, being based wholly on limited domains, language pairs, and certain test benchmarks, i.e., it lacks statistical significance power. Translations by neural MT tools like DeepL Translator, which is thought to usually deliver the best machine translation results as of 2022, typically still need post-editing by a human. Instead of training specialized translation models on parallel datasets, one can also directly prompt generative large language models like GPT to translate a text. This approach is considered promising, but is still more resource-intensive than specialized translation models.

3.

The rule-based machine translation approach was used mostly in the creation of dictionaries and grammar programs. Its biggest downfall was that everything had to be made explicit: orthographical variation and erroneous input must be made part of the source language analyser in order to cope with it, and lexical selection rules must be written for all instances of ambiguity.

4.

Transfer-based machine translation was similar to interlingual machine translation in that it created a translation from an intermediate representation that simulated the meaning of the original sentence. Unlike interlingual MT, it depended partially on the language pair involved in the translation.

5.

In the early 2000s, options for machine translation between spoken and signed languages were severely limited. It was a common belief that deaf individuals could use traditional translators. However, stress, intonation, pitch, and timing are conveyed much differently in spoken languages compared to signed languages. Therefore, a deaf individual may misinterpret or become confused about the meaning of written text that is based on a spoken language. Researchers Zhao, et al. (2000), developed a prototype called TEAM (translation from English to ASL by machine) that completed English to American Sign

Language (ASL) translations. The program would first analyze the syntactic, grammatical, and morphological aspects of the English text. Following this step, the program accessed a sign synthesizer, which acted as a dictionary for ASL. This synthesizer housed the process one must follow to complete ASL signs, as well as the meanings of these signs. Once the entire text is analyzed and the signs necessary to complete the translation are located in the synthesizer, a computer generated human appeared and would use ASL to sign the English text to the user.

6.

Non-standard speech One of the major pitfalls of MT is its inability to translate non-standard language with the same accuracy as standard language. Heuristic or statistical based MT takes input from various sources in standard form of a language. Rule-based translation, by nature, does not include common non-standard usages. This causes errors in translation from a vernacular source or into colloquial language. Limitations on translation from casual speech present issues in the use of machine translation in mobile devices.

7.

Copyright Only works that are original are subject to copyright protection, so some scholars claim that machine translation results are not entitled to copyright protection because MT does not involve creativity. The copyright at issue is for a derivative work; the author of the original work in the original language does not lose his rights when a work is translated: a translator must have permission to publish a translation.

8.

Applications While no system provides the ideal of fully automatic high-quality machine translation of unrestricted text, many fully automated systems produce reasonable output. The quality of machine translation is substantially improved if the domain is restricted and controlled. This enables using machine translation as a tool to speed up and simplify translations, as well as producing flawed but useful low-cost or ad-hoc translations.

9.

Studies using human evaluation (e.g. by professional literary translators or human readers) have systematically identified various issues with the latest advanced MT outputs. Some quality evaluation studies have found that, in several languages, human translations outperform ChatGPT-produced translations in terminological accuracy and clarity of expression. Common issues include the translation of ambiguous parts whose correct translation requires common sense-like semantic language processing or context. There can also be errors in the source texts, missing high-quality training data and the severity of frequency of several types of problems may not get reduced with techniques used to date, requiring some level of human active participation.

10.

Travel Machine translation applications have also been released for most mobile devices, including mobile telephones, pocket PCs, PDAs, etc. Due to their portability, such instruments have come to be designated as mobile translation tools enabling mobile business networking between partners speaking different languages, or facilitating both foreign language learning and unaccompanied traveling to

foreign countries without the need of the intermediation of a human translator. For example, the Google Translate app allows foreigners to quickly translate text in their surrounding via augmented reality using the smartphone camera that overlays the translated text onto the text. It can also recognize speech and then translate it.

11.

Medicine Despite being labelled as an unworthy competitor to human translation in 1966 by the Automated Language Processing Advisory Committee put together by the United States government, the quality of machine translation has now been improved to such levels that its application in online collaboration and in the medical field are being investigated. The application of this technology in medical settings where human translators are absent is another topic of research, but difficulties arise due to the importance of accurate translations in medical diagnoses. Researchers caution that the use of machine translation in medicine could risk mistranslations that can be dangerous in critical situations. Machine translation can make it easier for doctors to communicate with their patients in day-to-day activities, but it is recommended to only use machine translation when there is no other alternative, and that translated medical texts should be reviewed by human translators for accuracy.